MAHATMA GANDHIUNIVERSITY,KOTTAYAM

MGU-UGP(HONOURS)REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATIONNOVEMBER2025

FIRSTSEMESTER

Multi-DisciplinaryCourses(MDC)-MG1MDCFSQ100-FOODPROCESSING TECHNOLOGY

(2024ADMISSIONONWARDS)

Duration:2Hours **MaximumMarks:35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PartA

MultipleChoiceQuestions

Answerallquestions

1. d). Cutting & Chopping
2. a). Fermentation
3. d). Killing pathogenic microorganisms, such as bacteria and viruses.
4. c). Pasteurization
5. b). Rotary dryer
6. a). fermentation
7. a). Change in colour or odour
8. d). Kill pathogens while retaining product quality
9. a). 4°C
10. c). Freezing
11. d). Phosphatase
12. b).Radiation
13. d). May cause the formation of harmful compounds
14. a). High acid foods can be processed at 100°C
15. d). all of the above

PartB

ShortEssaytypeQuestions

Answer4questions

Eachquestioncarries5marks

16. Comment on nutrient losses during food processing

Nutrient losses during food processing occur due to exposure to heat, light, oxygen, pH changes and mechanical operations. Heat-sensitive vitamins such as vitamin C, thiamine and folate are most vulnerable during thermal processing methods like blanching, boiling and canning. Water-soluble nutrients also leach out into the cooking water when foods are washed, soaked or boiled. Oxidation during cutting, grinding and prolonged storage further reduces the levels of vitamins A, E and certain antioxidants. However, some processing methods such as steaming, minimal processing and vacuum cooking help retain nutrients more effectively. Although some nutrient loss is unavoidable, processing can improve the bioavailability of nutrients in certain foods, such as increasing lycopene availability in cooked tomatoes. Thus, nutrient loss depends on the type and intensity of the processing method used.

17. Describe the differences between tertiary processing and primary and secondary processing

Primary processing involves basic operations that convert raw agricultural materials into forms suitable for consumption or further processing, such as cleaning, sorting, milling, peeling and drying. Secondary processing transforms these primary products into more refined or finished foods, such as baking bread from flour, making cheese from milk or producing jams and sauces. Tertiary processing, however, deals with the creation of highly processed, ready-to-eat or convenience foods designed for immediate consumption. Examples include packaged snacks, instant noodles, ready-to-heat meals, breakfast cereals and frozen dinners. While primary and secondary processing focus on raw material transformation and product development, tertiary processing emphasizes convenience, extended shelf life, value addition and consumer appeal through advanced food engineering and packaging technologies.

18. Write a short note on the importance of food processing

Food processing plays a crucial role in ensuring food safety, improving shelf life and providing convenience to consumers. It helps in destroying harmful microorganisms, preventing spoilage and reducing foodborne illnesses. Processing methods such as pasteurization, canning, freezing and dehydration allow foods to remain stable for long periods, reducing post-harvest losses. It also improves the digestibility, palatability and nutritional value of foods through controlled heating, fermentation and fortification. Furthermore, processing creates a wide variety of food products that cater to diverse dietary needs, such as low-fat, fortified, gluten-free or ready-to-eat

foods. By enhancing market value, reducing wastage and enabling year-round availability of seasonal products, food processing becomes an essential part of modern food systems.

19. Discuss the advantages and disadvantages of drying as a food preservation method

Drying offers several advantages in food preservation. It significantly reduces the moisture content, which prevents the growth of bacteria, yeasts and molds. This extends shelf life and reduces the need for refrigeration. Dried foods are lightweight, easy to store and convenient for transport. Drying also retains many nutrients compared to severe heat treatments. However, drying has some disadvantages. Heat-sensitive vitamins like vitamin C and B-complex can be lost during drying. Some drying methods may cause shrinkage, texture changes and loss of natural flavor. Improper drying or storage may lead to moisture absorption and spoilage. Additionally, certain traditional drying methods depend heavily on weather and may not ensure uniform drying. Despite limitations, drying remains an economical and effective preservation technique.

20. Give a detailed note on dehydration with its principle

Dehydration is a food preservation technique that involves removing moisture from food to inhibit microbial growth and enzymatic activity. The principle of dehydration is based on reducing water activity (aw), thereby preventing microorganisms from multiplying. In dehydration, heat is applied under controlled conditions to evaporate water from the food surface, while air circulation carries the moisture away. Modern dehydration methods include tray drying, spray drying, drum drying, freeze drying and tunnel drying. Dehydrated foods become lighter, shelf-stable and easy to store and transport. The process must be carefully controlled to prevent case hardening, nutritional loss and undesirable texture changes. By maintaining optimal temperature, airflow and humidity, dehydration preserves the nutritional quality, color and flavor of foods more effectively.

21. Explain canning of fruits

Canning of fruits is a heat-based preservation method in which fruits are processed and sealed in airtight containers to prevent spoilage. First, fresh fruits are washed, peeled, trimmed and cut as required. They are then filled into cans or jars along with syrup, juice or water to improve flavor and texture. The filled containers are exhausted to remove trapped air, which helps create a vacuum seal. Next, the cans are sealed and subjected to heat treatment at appropriate temperatures to destroy spoilage organisms and enzymes. Since fruits are high-acid foods, they can be safely processed at around 100°C in a water bath. After heating, the cans are rapidly cooled to prevent overcooking and ensure product stability. Proper canning maintains the color, flavor and nutritional quality of fruits while extending shelf life for months or years.